



Communication

Estimate bond angle dependence of superconducting transition temperature in NaFeAs with the first principle methods

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ABSTRACT

The change of superconducting transition temperature (T_c) with bond angle in iron-pnictides was investigated by first-principles calculation based on density functional theory. A Green's function method was adopted to estimate the maximum eigenvalue of Eliashberg equation (an indicator of T_c) for NaFeAs with different bond angles. Through calculations the band structure of NaFeAs was obtained. It was found that hole pockets could form at the center of the first Brillouin zone in the band structure, which confirmed the existence of superconductivity in NaFeAs. The upper limit of the Fe–As–Fe bond angle for superconductivity was found to be 121° and the highest T_c would occur at the angle of 108° .

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1. Introduction

Superconductivity was first discovered in metals, then metallic oxides [1–3] and, recently, iron based superconductors [4,5]. The BCS model (by Bardeen, Cooper, and Schrieffer) was the first comprehensive theory on superconductivity [6], where the Cooper pair, formed by two electrons near the Fermi level through phonon interactions, is thought to be the key to superconductivity. The critical transition temperature (T_c) is usually estimated with the McMillan–Allen–Dynes equation [7,8]. However, the BCS theory is not applicable to the iron based superconductors.

To estimate the T_c of iron-based superconductors, Kuroki et al. developed a series of methods which is based on the Green's functions. In their methods, the band structure is first calculated with local density approximation (LDA), then maximum localized Wannier functions (MLWFs) are constructed. At last, Green's functions can be calculated with random phase approximation (RPA) [9,10] or fluctuation-exchange approximation (FLEX), [11–

14] with the results being plugged into the Eliashberg equation [15] to obtain the eigenvalue, [16,17] an indicator of T_c .

Analogue to that in many other functional materials (e.g. ferroelectric oxides, colossal magnetoresistance compounds), [18–22] the electrical/magnetic properties of iron-based superconductors are sensitive to lattice distortions which are often present in thin film geometries. In particular, the Fe–Pn–Fe (Pn refers to pnictogen) bond angle is a key factor to superconductivity in iron-based materials [23]. Despite numerous experimental data in the literature, a theoretical study on the relation between T_c and bond angle is still lacking. In this paper, since it can be easily fabricated into sizable and high-quality single crystals, [24–26] we choose NaFeAs as the model superconductor to study how the Fe–As–Fe bond angle affects the T_c with density functional theory (DFT) and Green's function methods. As to be demonstrated later, there exists an upper limit of the Fe–As–Fe angle for the NaFeAs to exhibit superconductivity as well as an optimal value of the angle for the highest T_c in this compound.

2. Calculation methods

2.1. LDA method

The unit cell of NaFeAs is shown in Fig. 1(a), in which the red

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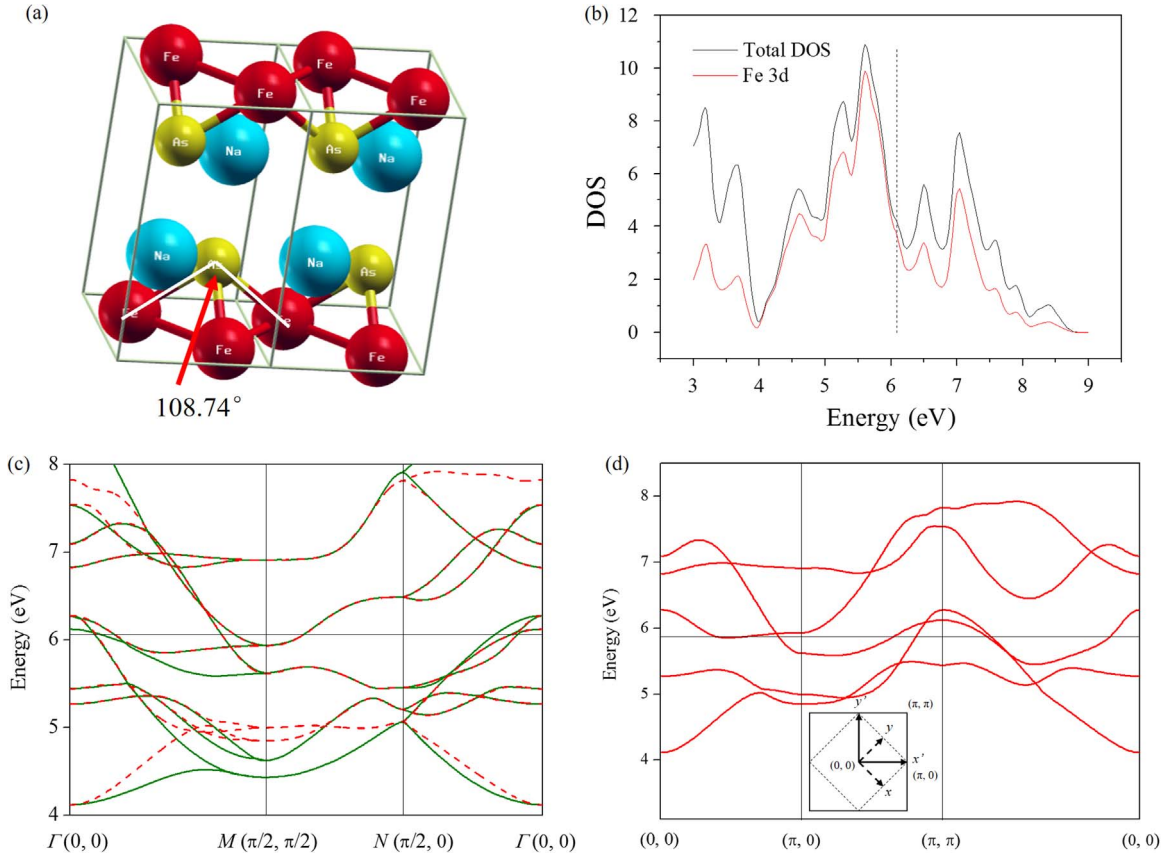


Fig. 1. (Color online) (a) The unit cell of NaFeAs. The Fe–As–Fe bond angle is marked by solid white lines. (b) The DOS and PDOS of Fe-3d near the Fermi surface (dashed line). (c) The LDA energy bands (green solid lines) and the MLWF bands constructed with the Fe-3d orbitals (red dashed lines). (d) The 5 MLWF bands in the unfolded BZ, which are transferred from the 10 MLWF bands.

atoms are Fe, the yellow atoms are As, and the blue atoms are Na [27]. The initial lattice parameters for calculations are taken from Ref. [27] at 2.5 K: $a=b=3.9507 \text{ \AA}$, $c=7.0148 \text{ \AA}$. The Fe–As–Fe bond angle, which is 108.74° , has been marked in the Fig. 1(a). To exclude the possible influence of other factors when changing the bond angle, we fix the distance between the two Na atom layers, the relative positions between As and Na atoms, and the Fe–As bond length [28]. The a , b , and c can be expressed as functions of bond angle.

$$\begin{aligned} a &= 2\sqrt{2}l_{\text{Fe-As}} \sin(\theta/2) \\ b &= a \\ c &= l_{\text{As-As}} + 2l_{\text{Fe-As}} \cos(\theta/2) \end{aligned} \quad (1)$$

where θ means the Fe–As–Fe bond angle, $l_{\text{Fe-As}}$ means the Fe–As bond length, and $l_{\text{As-As}}$ means the vertical distance between the 2 As atoms.

The band structure and Fermi surfaces of NaFeAs were calculated by local density approximation (LDA) using Quantum ESPRESSO codes [29]. In the calculations, the $\mathbf{k}$ -point grid is $8 \times 8 \times 8$, the kinetic energy cutoff for wave functions is 50 Ryd, and the kinetic energy cutoff for charge density is 200 Ryd.

2.2. MLWF method

Due to the special DOS and band structures (will be discussed in Section 3.1), we construct the maximum localized Wannier functions (MLWFs) using the Fe-3d orbitals as the bases with the WANNIER90 code [31]. Furthermore, we can simplify the 10 bands into 5 bands in the unfolded BZ, with supposing the basis vectors of the 1st BZ and the unfolded BZ be x, y and x', y' (inset of Fig. 1

(d)), respectively [10,16,17,30,31].

2.3. FLEX method

To explore the relation between T_c and bond angle, we use fluctuation-exchange approximation (FLEX) [11,16] to calculate many-body Hamiltonian model. We add the following one-site Coulomb interaction terms to the Hamiltonian: [16]

$$\begin{aligned} H' = & \frac{U}{2} \sum_{il} \sum_{\sigma} c_{il\sigma}^\dagger c_{il\sigma}^\dagger c_{il\sigma} c_{il\sigma} \\ & + \frac{U'}{2} \sum_{il \neq m} \sum_{\sigma\sigma'} c_{il\sigma}^\dagger c_{im\sigma'}^\dagger c_{im\sigma} c_{il\sigma} \\ & + \frac{J}{2} \sum_{il \neq m} \sum_{\sigma\sigma'} c_{il\sigma}^\dagger c_{im\sigma}^\dagger c_{il\sigma'} c_{im\sigma'} \\ & + \frac{J'}{2} \sum_{il \neq m} \sum_{\sigma\sigma'} c_{il\sigma}^\dagger c_{il\sigma}^\dagger c_{im\sigma} c_{im\sigma} \end{aligned} \quad (2)$$

where i means a site, l, m means orbitals, $\sigma = \pm$, $\bar{\sigma} = -\sigma$, $c_{i\sigma}^\dagger$ means creation operator and $c_{i\sigma}$ means annihilation operator in the basis of real harmonics at Fe sites, U is the intraorbital, U' is the interorbital, J is the Hund's coupling, and J' is the pair hopping interaction. To meet the rotation invariance of the atomic orbitals in the orbital space, we take $U=U' + 2J$ and $J=J'$. In the calculation, the values of them are: $U=1.5 \text{ eV}$, $U'=1.2 \text{ eV}$, and $J=J'=0.15 \text{ eV}$ [17].

The hopping matrix is obtained from the MLWF band structure. In FLEX, the self-energy is calculated with the susceptibilities which are the summary of bubble-and-ladder-type diagrams consisting of renormalized Green's functions. The renormalized

Green's functions are then determined self consistently from the Dyson's equation.

$$G(k) = G^0(k) + G^0(k) \sum (k) G(k) \quad (3)$$

The self-energy function Σ is:

$$\sum_{lm}(k) = \frac{T}{N} \sum_q V_{ll',mm'}(q) G_{l'm'}(k - q) \quad (4)$$

The subscripts mean the Einstein's summary, $k \equiv (\mathbf{k}, i\omega_n)$ and $q \equiv (\mathbf{q}, i\nu_n)$ mean dynamic moment and fermionic/bosonic Matsubara frequency. V is effective interaction. N is the number of $\mathbf{k}$ grid.

The convergent Green's function is plugged into the linearized Eliashberg equation

$$\Delta_{lm}(k) = -\lambda \sum_{k'} V_{ll',m'm'}^{(t)}(k - k') G_{l'l'}(k') G_{m'm'}(k') \Delta_{l'm'}(k'), \quad (5)$$

where

$$V^s(q) = U_{\uparrow\downarrow} + \frac{3}{2} U^s \chi^s(q) U^s - \frac{1}{2} U^c \chi^c(q) U^c \quad (6)$$

for spin-singlet and

$$V^t(q) = U_{\uparrow\uparrow} - \frac{1}{2} U^s \chi^s(q) U^s - \frac{1}{2} U^c \chi^c(q) U^c \quad (7)$$

for spin-triplet. χ^s, χ^c are susceptibilities for spin and charge, respectively, $U^{s,c} = U_{\uparrow\uparrow} \mp U_{\uparrow\downarrow}$ are bare vertices. $T = T_c$, when the eigenvalue λ reaches unity. The detailed equations of these physical quantities, including V and the construction of G^0 can be read in Ref. [16].

As the Fermi surfaces are all cylinder like, the calculation can be simplified into a 2D surface by taking $k_z = 0$. In our calculation, we take an $8 \times 8 \times 1$ mesh in the unfolded BZ and 1024 Matsubara frequencies. The calculated Green's functions and self-energy functions are expressed as (5×5) matrices, and the susceptibility functions, such as V and U , are expressed as $((5 \times 5) \times (5 \times 5))$ matrices.

3. Results and discussions

3.1. Band structures and Fermi surfaces

The density of states (DOS) and band structure of the initial lattice are shown in Fig. 1(b) and ((c), green solid lines), respectively. In Fig. 1(b), the summary of the projected DOS (PDOS) of the 10 Fe-3d orbitals is also plotted. It can be seen that most of the DOS near the Fermi surface is contributed by Fe-3d orbitals, so we can construct the band structure with MLWFs basing these

orbitals, and the MLWF band structure is shown in Fig. 1(c) in red dash lines. The band structures show a special feature: the number of bands in the region from M to N is half of that in the regions from Γ to M and from Γ to N , due to degeneracy, so we can simplify the 10 bands into 5 bands in the unfolded Brillouin zone (BZ).

The Fermi surfaces are shown in Fig. 2(a). Three concentric cylindrical hole-like Fermi surfaces are located at the center of the 1st BZ, and four electron-like Fermi surfaces are located at the corners of the 1st BZ. This can also be found in the band structures. The nesting of these cylindrical Fermi surfaces is often thought to be the decisive factors to induce superconductivity [30].

We chose some bond angles to show the change of the number of Fermi surfaces in the unfolded BZ, as shown in Fig. 2(b), (c) and (d). α and γ are hole-like Fermi surfaces and β is electron-like Fermi surface. Additionally, when the bond angle is greater than 111° , or less than 101° , the number of hole-like Fermi surfaces will be reduced from 3 to 2, implying the degradation of superconductivity. When the bond angle is greater than 121° (122° in Fig. 2(d)), the hole-like Fermi surface disappears, suggesting the disappearance of nesting and superconductivity.

3.2. FLEX results

The eigenvalues are calculated at a fixed temperature ($T = 0.01$ eV, about 58 K), the value of λ is then used as an indicator of T_c , with larger λ corresponding to higher T_c , [28] and the larger λ of spin-singlet and spin-triplet should be taken as the dominant state. Results are shown in Fig. 3, we found that spin-singlet always has much larger eigenvalue which means it holds the dominant position. We can also find λ reaches its maximum value at 108° which should correspond to the highest T_c , then decreases suddenly with increasing the bond angle into 109° . Our result is comparable with the experimental one, [24] where the maximum T_c of NaFeAs was found around a bond angle of 109° .

Some bond angles around the maximum eigenvalue were chosen to analysis the gap functions and pair susceptibility. We chose some bond angles near the highest T_c to make deeper analysis about the gap functions and spin susceptibilities. The gap functions for the forth band at $\tau = 0$ with bond angles of 105° , 108° and 110° are shown in Fig. 4. Because the gap function can be viewed as an eigenfunction of linear Eliashberg equation, the magnitude of that on each $\mathbf{k}$ point is of no meaning and we should only focus on the relative value, while blue regions for lower and yellow regions for higher. We find all the gap functions are of s-wave symmetric, which is comparable with the fitting results in Ref. [26] but the node distributions are very different. For 105° and 108° , there no nodes and the gap function are opened. The gap function of 110° has a nodal morphology, which will break the

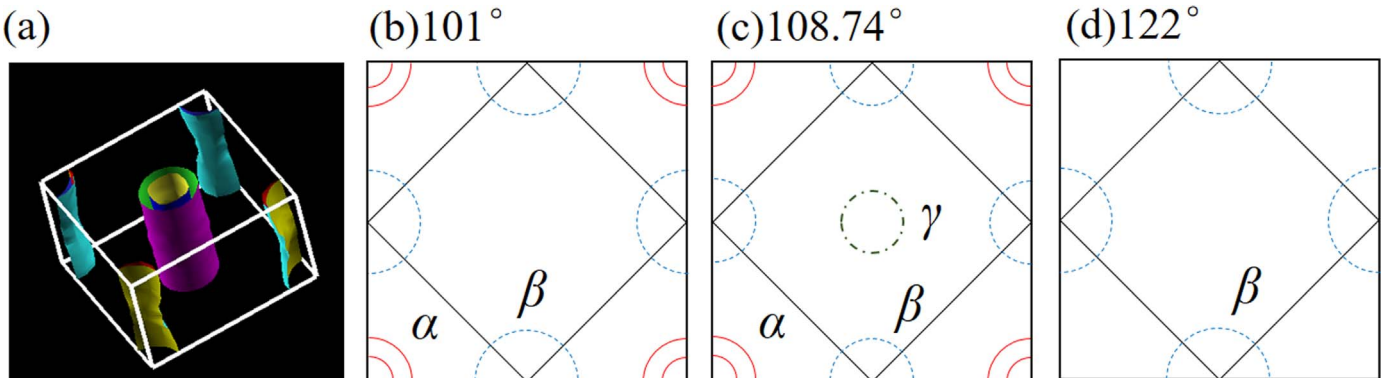


Fig. 2. (Color online) (a) The sketches of the Fermi surfaces in unfolded BZ, (b)–(d) The number of the hole-like Fermi surfaces v.s. bond angle. α and γ are both hole-like and β is electron-like.

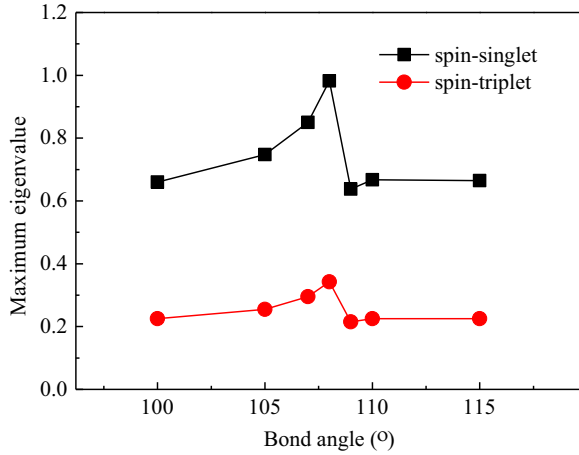


Fig. 3. (Color online) The maximum eigenvalue of linear Eliashberg equation v.s. bond angle.

electron pair and is not good for superconductivity, indicating a lower transfer temperature.

The spin susceptibilities and sketches of Fermi surfaces at $\nu=0$

of 105°, 108° and 110° are also show in Fig. 4. We find the spin susceptibilities are concentrate on $(\pi, 0)$ and its symmetrical points, which reflects the nesting between α/γ and β pockets. Though the topologies of them are very similar, the magnitudes of 105° and 108° is lower than that of 110°, which indicates 110° has a better magnetic stability. About the Fermi surfaces, the γ electron pocket is the largest at 108°, and the two α surfaces of 110° separate more obviously.

In a word, our results suggest the abrupt change of the eigenvalue is induced by an obvious nodal gap function form when the Fe–As–Fe bond angle is big enough.

4. Conclusions

Assuming that T_c increases with λ , the dependence of T_c on bond angle of iron-based superconductor NaFeAs is studied by Green's functions derived from a combined local density approximation, maximum localized Wannier functions and fluctuation-exchange approximation methods. Hole-like Fermi surfaces are found when the Fe–As–Fe bond angle is less than 121°, confirming the existence of superconductivity. From the form of gap functions, we think the superconductive pair mechanism of NaFeAs should

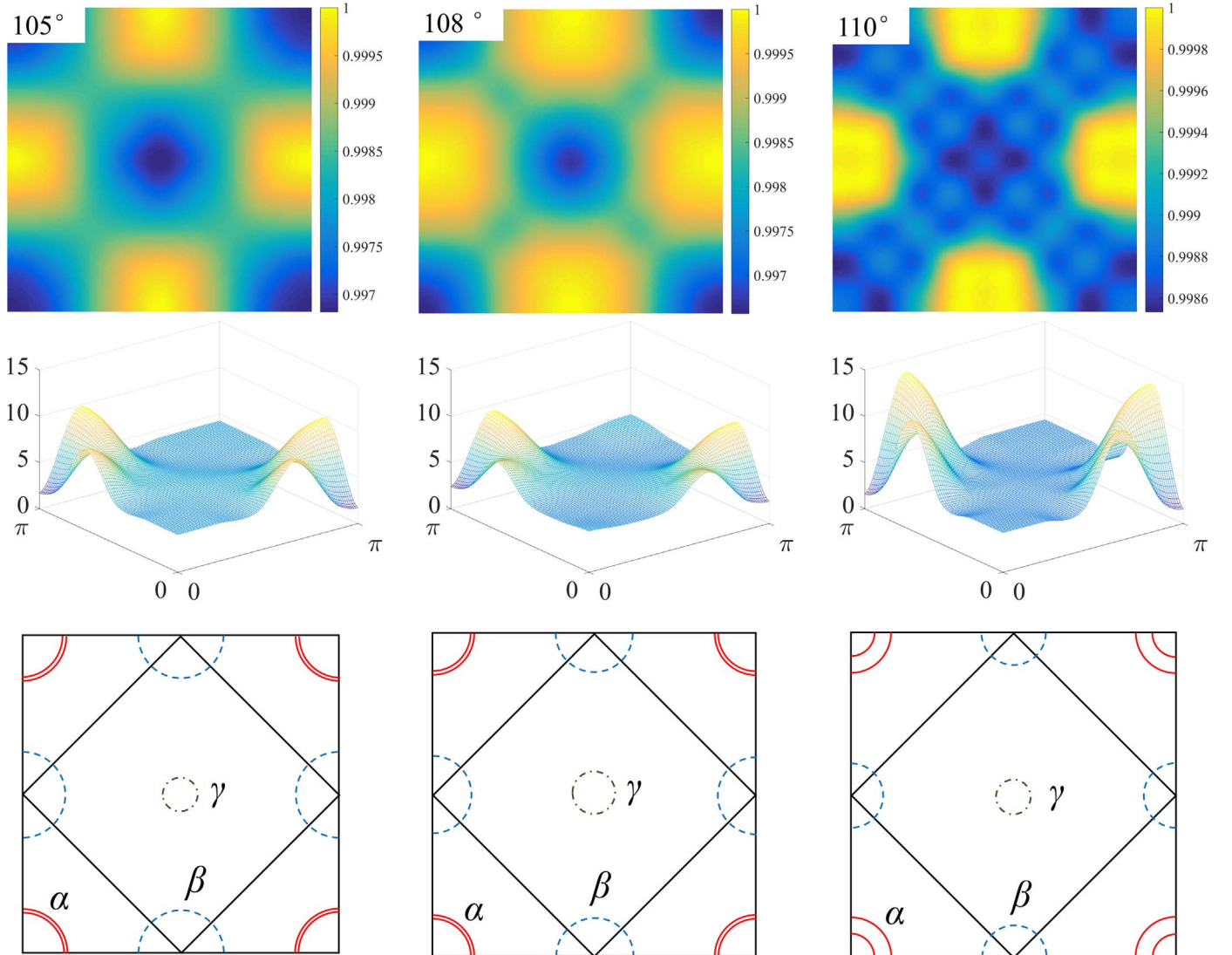


Fig. 4. (Color online) The patterns of gap functions (top), spin susceptibilities (middle) and sketches of Fermi surfaces (bottom) of 105°, 108° and 110° (from left to right, the regions of gap functions and Fermi surfaces are from $-\pi$ to π).

be of s -wave. From the maximum eigenvalues of linear Eliashberg equation, it was found that T_c changes with the bond angle and the maximum T_c occurred around a bond angle of 108° .

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